

Aaryan Nakhat

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SUMMARY

Senior Manager, AI/ML with 3+ years building **production LLM and speech systems** end-to-end: **fine-tuning** and **preference-tuning** (SFT, DPO), **multilingual streaming speech-to-text fine-tuning**, **retrieval from vector search** to **GraphRAG**, and **real-time multi-agent voice** infrastructure, with measured production impact (**50% lower on-call latency**, **85% lower operating cost**). Also ship substantial independent systems from scratch: a primary-source **equity-research & forensic-analysis workbench** for Indian markets and **CricDex**, an open cricket-intelligence platform. Both span anti-bot data ingestion, deterministic and Bayesian modeling, LLM reasoning, and always-on delivery.

EXPERIENCE

SalesAgents AI

Senior Manager, AI/ML

AI/ML Engineer

Mohali, India

Aug 2026 – Present

Sep 2023 – Jul 2026

- **Data & evaluation infrastructure:** Built robust data pipelines for prep, cleaning, transformation, and synthetic-data generation, plus **continuous-evaluation and automated-improvement pipelines** that cut manual effort in the eval-and-improvement loop by **75%**
- **LLM fine-tuning:** **Fine-tuned** multiple small **LLMs** for diverse use cases including conversational AI, bidirectional translation, structured information extraction, topic modeling, action prediction, and custom guardrails
- **Preference alignment:** **Preference-tuned** the fine-tuned LLMs using techniques like **DPO** to better align model outputs with user preferences and improve response quality
- **RAG from scratch:** Iteratively built a **RAG system** starting with **vector-similarity** retrieval (semantic/agentic chunking, re-ranking, query rewriting, HyDE), evolving to **Neo4j** knowledge graphs, and reaching best performance with **GraphRAG** (LightRAG)
- **Voice-agent backend:** Developed and scaled **AI backend systems** on LiveKit with **multi-agent orchestration**, **tool calling**, **caching**, **load balancing**, and **observability** for production-grade voice agents
- **Real-time optimization:** Optimized real-time voice-agent infrastructure, **reducing on-call latency by 50%** and **cutting operational cost by 85%** through efficient system design and resource utilization
- **Voice ordering agent (8,000+ product catalog):** Built a system that lets dealers place large orders over a call, matching what they say to the right item through **layered, hierarchical search** over an **Elasticsearch** catalog (product → shade → pack), with a recovery step that **re-listens to the call audio to fix speech-to-text errors** on mangled names; live stock and pricing come from the client's systems via **tool calls**, and pack sizes and discounts are computed by **dedicated tools the model calls**
- **Streaming ASR fine-tuning:** Fine-tuned our own **real-time streaming speech-to-text model** (**cache-aware FastConformer** encoder, **TDT** decoder) for **9 Indian languages**, handling **mixed-language speech**, with **context biasing** to boost product names and **end-of-turn detection**

ValueLabs

Software Development Intern

Hyderabad, India

Jan 2023 – Sep 2023

- **Domain LLM in production:** Developed a custom **fine-tuned insurance chat expert**, deployed on a large insurance company's website
- **Guardrails:** Designed **custom guardrails** on top of the fine-tuned LLM for **input and output control**
- **Goal-directed control:** Created an additional **auxiliary classifier** to guide the LLM toward its desired goal at each step

EDUCATION

ICFAI Foundation for Higher Education

Bachelor of Technology in Data Science and Artificial Intelligence; CGPA: 9.78/10

Hyderabad, India

Aug 2019 – Jun 2023

- Achievements: **gold medal** for all-round excellence (best student), **silver medal** for academic excellence, **100% scholarship in all 8 semesters**, won 2 intra-college coding competitions
- Extracurricular activities: vice president of the technical club, part of the university's cricket team
- Relevant Coursework: Probability, Statistics, Data Science, Machine Learning, Deep Learning, Data Visualization, Data Mining, Data Warehousing, Blockchain Technology

PROJECTS

Personal Equity-Research & Forensic Analysis Workbench (Indian Markets) (*code*) Jun 2026 – Present

- **Primary-source data engine: Anti-bot scrapers** pull exchange filings, financials, live prices, derivatives, money-flow, commodity and mutual-fund data, rating and promoter filings, IPO prospectuses, US FDA filings and vehicle registrations into a **40+ table DuckDB** database adjusted for every split, bonus and demerger
- **Forensic & valuation core:** Multi-year financials and cash-flow quality, a **forensic block** that flags accounting manipulation and bankruptcy risk, and **sector-aware valuation** with a **Monte-Carlo** fair-value model and a reverse model that back-solves the growth the current price implies
- **Ownership & smart-money:** Tracks every holder and how big investors add or exit **quarter to quarter**, plus insider trades, pledges and block deals, and estimates where institutions bought to flag **profit-booking risk**
- **Idea-discovery & signal engines: 24 screens and 7 radars** find new stock ideas, and a **Hotlist** ranks the names several of them agree on. Highlights:
 - **Value idea-finder:** ranks the **top 500 stocks** on quality, forensic health and cheapness vs their own history
 - **Top-down sector engine:** for each sector index, trend, **valuation vs its own 5-yr history**, smart-money accumulation, and the best & cheapest names inside it, with a weekly **sector-rotation** leaders/laggards read
 - **Global supply-shock finder:** a **four-stage multi-agent pipeline** (scout → triage → map → audit) that turns export bans / quotas / tariffs into **verified** Indian beneficiaries, each stock checked against a master list
 - **Demand-surge finder:** agents map rising demand (search trends + news) to **listed suppliers**
 - **Earnings-call check:** an **LLM compares management's promises** on calls with **results delivered**
 - **Technical-setup screener:** best chart setups with entry / stop / target, filtered on forensic health
 - **Holding-company discount finder:** holding companies trading **below the value of the stakes they own**
 - **Marquee-investor tracker:** what ~25 well-known investors bought, added, trimmed or sold last quarter
 - **Capex-led small-cap hunter:** small companies **investing heavily in new capacity**, minus risky ones
 - **Government policy radar:** maps each **government scheme** to the **companies** likely to benefit
 - **Filing-reading signals:** an **LLM reads** rating, promoter-issue, management-change and business-sale filings, each fact **checked word for word against the source** (upgrades, owners buying above market, new CEOs)
 - **Alternative-data early signals:** reads non-market data to **spot changes before results**, using US FDA filings for Indian pharma, vehicle registrations, UN import data, raw-material prices
- **Mutual-fund & IPO coverage:** Deep reports on ~14,500 mutual-fund schemes (returns, risk, benchmark) and pre-listing **IPO** notes weighing pricing, demand and risks into an **apply-or-avoid** call
- **Investor tools: Tax-aware portfolio** tracking that applies every split, merger and demerger, a **fact-checker for stock tips** seen on social media, and a **track record** that scores every call against the market
- **Investment thesis & always-on delivery:** An **LLM** writes a thesis with **upside drivers** and a **sell-priority** guide, delivered via a **local web app** (50 tools), a **CLI** and an **email bot** as **PDF** reports and daily digests

CricDex: Open Cricket Intelligence Platform (*demo*) (*code*) May 2026 – Jun 2026

- **Bayesian skill engine:** Rated players with dismissal-aware, opponent-adjusted **Bayesian** models (**JAX/NumPyro**, **ADVI/NUTS**), scoring batters on run-getting and survival and bowlers on economy and strike rate, trained on **Cricsheet** ball-by-ball data held in **DuckDB**
- **Novel cricket metrics:** Designed 10 context-adjusted metrics that measure match impact better than batting average or economy, including **Net Game Impact** (per-ball win-probability added, via an **XGBoost** model), **Wicket Quality**, **Crease Longevity**, **Slow-Start Cost**, etc.
- **Scout & auction simulator:** Built a cross-league **Scout** that finds player look-alikes across 6 T20 leagues using a **Bayesian** skill score, each with an estimated auction price, and a rules-accurate **IPL auction simulator** (**Monte-Carlo**, ~300 runs) supporting real retentions, overseas caps, and second-price bidding

CERTIFICATIONS AND SKILLS

LLM & Agents: LLM fine-tuning (SFT, DPO), RAG → GraphRAG (LightRAG), LangChain, LiveKit multi-agent orchestration, tool-calling, semantic/agentic chunking, re-ranking, query rewriting, HyDE, custom guardrails

Retrieval & Databases: Neo4j knowledge graphs, vector databases, Elasticsearch, DuckDB, SQL

ML & Serving: PyTorch, NVIDIA NeMo, JAX/NumPyro, vLLM, SGLang, Docker, Kubernetes, FastAPI

Data & BI: Tableau, Power BI

Cloud & Tools: AWS, GCP, Azure, Linux, Git, VS Code, PyCharm, Jupyter Notebooks

Languages: Python, C, C++, SQL

Certifications (clickable links): Python; Data Science and Machine Learning; Deep Learning; Custom Models, Layers and Loss Functions; Building LLM-Powered Applications; REST APIs; SQL